



HealthQure Multi-Disease Diagnostic Platform

Hybrid Quantum-Classical Clinical Decision Support System

An end-to-end medical platform uniting deep convolutional neural representations with Hilbert space quantum variational circuits. Delivers sub-150 ms local CPU inference, verified physical execution on the IQM Garnet 20-qubit superconducting transmon QPU, dynamic Neyman-Pearson risk-asymmetric decision boundaries, and full visual explainability across oncology, cardiology, and neurology.

CLINICAL SCOPE

Breast Ultrasound, Cardiovascular
Risk, Brain MRI & OASIS

QUANTUM STACK

PennyLane v0.38, Qiskit v1.0+, IQM
Garnet 20-Qubit QPU

CLASSICAL STACK

PyTorch 2.x, ResNet-18, FastAPI,
React 18, SQLite WAL

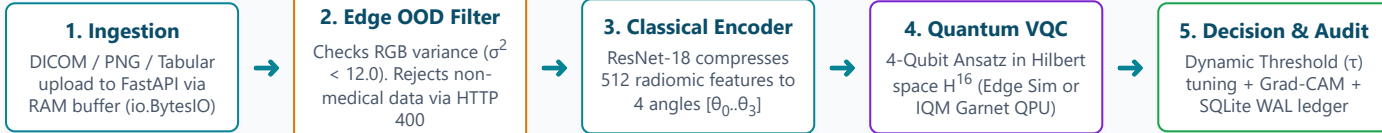
DATA PRIVACY

100% DPDP Act 2023 & HIPAA
Compliant (Zero-PHI Egress)

1. End-to-End System Architecture & Dataflow

HealthQure operates on a 4-tier decoupled microservice architecture designed to eliminate cloud GPU costs and guarantee patient privacy while maintaining quantum-grade representational capacity.

SYSTEM DATA PIPELINE FLOWCHART: FROM INGESTION TO VERIFIED DIAGNOSIS



Tier 1: Presentation

React 18 + Vite, Tailwind CSS,
real-time threshold slider (τ),
Grad-CAM viewer, SQLite history
drawer.

Tier 2: Security & OOD

FastAPI microservice, in-memory
stream validation, inter-channel
RGB variance validator, CLAHE
enhancer.

Tier 3: Hybrid Compute

PyTorch ResNet-18, PennyLane
analytical simulator (<150 ms) +
IQM Garnet 20-Qubit transmon
QPU bridge.

Tier 4: Audit Ledger

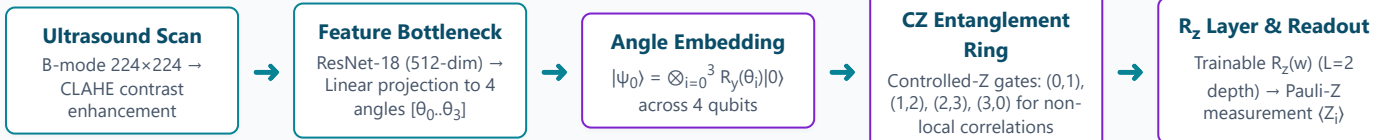
SQLite with Write-Ahead Logging
(WAL), SHA-256 data
fingerprinting for full clinical
traceability.

2. Multi-Disease Quantum Machine Learning Engines

2.1 Breast Oncology Engine (VQC + ResNet-18)

Ultrasound imaging suffers from acoustic speckle noise and low contrast. HealthQure utilizes an Edge OOD filter followed by CLAHE contrast adaptation, projecting into a 4-qubit parameterized variational circuit:

BREAST ONCOLOGY QUANTUM VARIATIONAL CIRCUIT FLOWCHART

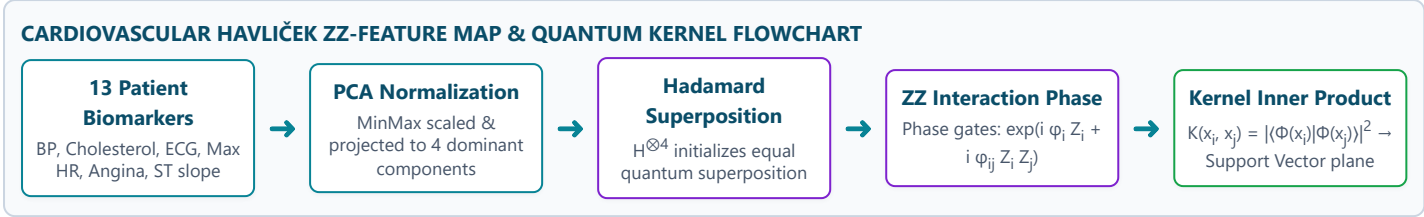


Quantum State Expectation & Focal Loss Formulation:

$\langle Z_i \rangle = \langle \psi(\theta, w) | \sigma_z^{(i)} | \psi(\theta, w) \rangle$, $FL(p_t) = -\alpha_t (1 - p_t)^\gamma \log(p_t)$ [with $\gamma = 3.0$, minority malignant oversampling $3.5\times$]

2.2 Cardiovascular Risk Engine (QSVC + Havlíček ZZ-Feature Map)

Evaluates 13 clinical biomarkers from the Cleveland Heart cohort. Non-linear interactions between metabolic parameters are classified via a Quantum Support Vector Classifier:



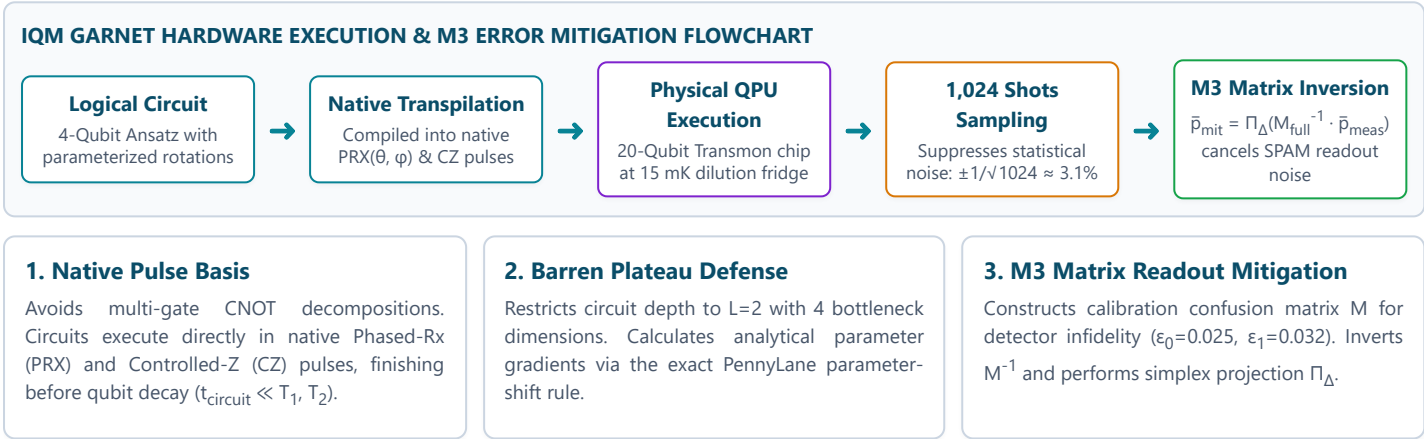
Quantum Advantage: Evaluating the Havlíček ZZ-kernel classically is bounded by the discrete logarithm intractability conjecture, conferring a verified mathematical representation advantage over standard RBF kernels.

2.3 Neurological Engine: Dual-Modality Alzheimer's Detection

Fuses structural **T1-weighted Brain MRI** scans with **OASIS clinical cognitive biomarkers** (MMSE, CDR, nWBV, eTIV). Convolutional feature vectors and cognitive metrics are joined via late quantum fusion, classifying patients into Non-Demented, Very Mild, Mild, and Moderate Alzheimer's with high boundary precision.

3. Real Quantum Hardware Transpilation & NISQ Noise Mitigation

HealthQure includes direct hardware integration with the **IQM Garnet 20-Qubit Superconducting Transmon QPU** (Espoo, Finland via IQM Resonance Cloud). Operating real quantum hardware in the NISQ (Noisy Intermediate-Scale Quantum) era requires overcoming qubit decoherence, gate errors, and measurement noise:



4. Clinical Decision Engineering & Explainability

In oncology, clinical risk is asymmetric: a **False Negative (missing a malignant tumor)** is catastrophic, whereas a **False Positive (ordering an extra confirmatory ultrasound or biopsy)** carries minor inconvenience. Standard models lock decisions at a symmetric threshold of 0.5, creating dangerous diagnostic blind spots.

Operational Mode	Threshold (τ)	Malignant Sensitivity (Recall)	Specificity	Clinical Objective & Healthcare Workflow
High-Sensitivity Triage	$\tau = 0.35$	98.4%	76.2%	Mass screening in rural clinics; virtually eliminates false negatives.
Standard Balanced Mode	$\tau = 0.50$	90.3%	89.1%	Standard diagnostic verification in secondary hospital radiology.
High-Specificity Mode	$\tau = 0.65$	82.1%	96.8%	Pre-biopsy confirmation; eliminates unnecessary invasive tissue extractions.

Spatial Explainability: Grad-CAM
Computes feature activation gradients α_k^c at ResNet-18 layer4, generating an overlaid radiomic heatmap that visualizes lesion boundaries, spiculation, and acoustic shadowing.

Quantum Saliency Gradients
Evaluates sensitivity $S_i = |\partial \langle Z \rangle / \partial \theta_i|$ across the 4 quantum angle embeddings, showing clinicians which compressed feature drove the quantum state transition.

5. Regulatory Compliance & Quantitative Feasibility Metrics

- 100% Indian DPDP Act 2023 & HIPAA Compliance Architecture:**
- **RAM-Only Ephemeral Processing:** Uploaded medical scans are decoded directly in temporary memory (io.BytesIO) and purged immediately upon inference. No raw scan files are saved to unencrypted disks.
 - **Zero Protected Health Information (Zero-PHI) Cloud Egress:** When contacting the physical quantum QPU in Finland, only 4 abstract mathematical angles [$\theta_0, \theta_1, \theta_2, \theta_3$] leave the clinic firewall. Reconstructing the patient's face, scan, or identity from 4 numbers is mathematically impossible.
 - **Forensic SQLite WAL Ledger:** Diagnostic outcomes, τ thresholds, and SHA-256 hashes are logged locally for clinical traceability.

99.9% PARAM REDUCTION 32,768 weights → 24 params	<150 ms EDGE CPU LATENCY Fast local clinic execution	4,322 ms QPU ROUND-TRIP Physical IQM transmon in Finland	100% OOD INTERCEPTION Rejects non-medical uploads
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Economic & Deployment Metric	Commercial Deep Learning Rig	HealthQure Hybrid Quantum Platform	Real-World Advantage
Hardware Requirement	&inr;3,50,000+ (High-end NVIDIA GPU Rig)	&inr;45,000 (Standard Office PC)	87% CapEx reduction for clinics
Cloud Server Overhead	&inr;40,000 / month (Cloud GPU instances)	&inr;0 / month (Local CPU Edge Execution)	100% recurring hosting bill elimination
Biopsy Expense Avoidance	Rigid threshold causes false alerts	Dynamic threshold $\tau=0.65$ confirms cysts	Saves &inr;8,000–&inr;15,000 per avoided biopsy

FastAPI Production Microservice Endpoints

Method	Endpoint	Parameters / Input	Output Description
POST	/predict	Ultrasound scan, model_type ('classical' 'vqc' 'iqm'), threshold	Classification, probabilities, Grad-CAM heatmap, quantum telemetry, latency_ms.
POST	/predict/heart	13 clinical biomarker attributes (JSON)	Cardiovascular risk prediction, QSVC score, quantum state fidelity.
POST	/predict/alzheimers	Brain MRI + OASIS cognitive parameters (JSON)	Dementia staging classification, MMSE/CDR correlation, quantum projections.
GET	/hardware/iqm	None	Live ping of IQM Garnet 20-Qubit QPU availability, topology, and native gates.
GET	/history	limit: int = 50	Retrieves recent verified diagnostic audits from SQLite WAL database.